

Scalability & Future Plan

Date	6 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the **OptiCrop – AI-Based Crop Recommendation System** can be enhanced to support a larger number of users, improve prediction accuracy, strengthen system performance, and introduce advanced smart farming features in future versions. It also presents the roadmap for expanding the application's capabilities beyond the current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No user authentication system	Anyone can access the application without login.	High
2	No database to store crop recommendation history	Previous recommendations cannot be saved or viewed later.	Medium
3	Uses only soil and weather parameters	Additional agricultural factors can improve recommendation accuracy.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on a scalable cloud platform with load balancing and auto scaling.
2	Data Storage	No database integration	Integrate MySQL or PostgreSQL to store user profiles and recommendation history.
3	Performance	Single Flask application	Optimize model loading, caching, and backend processing for faster predictions.
4	Security	Basic crop recommendation	Integrate real-time weather API, fertilizer recommendation, and irrigation suggestions.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add user authentication and farmer profile management	2–3 Months	Secure access and personalized recommendations.
Phase 3	Store crop recommendation history using a database	3–4 Months	Enable farmers to view and manage previous recommendations..
Phase 4	Integrate real-time weather forecasting, fertilizer recommendation, and advanced Machine Learning models	5–6 Months	Improve recommendation accuracy and provide complete smart farming assistance.